



Read the Turtle Code

Quick facilitation notes & answer key

Goal

Students read turtle for-loops and PREDICT the drawing before running (C3.2): a turn → a shape, no turn → a line, and the range number gives the sides. They also reason about where Bit ends.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

Python Turtle projected (or a web playground); the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Model reading Loop A aloud: 'turn → shape; range(4) → 4 sides → a square.'
2. Have students predict BEFORE you run each loop; then run to confirm.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: read the clues (turn? how many sides?).

Answer key (checked by the run-gate)

Ex 1 — a square; 4 sides.

Ex 2 — a triangle; 3 sides.

Ex 3 — a straight line; 100 long (5×20).

Ex 4 (challenge) — back at the start (a closed square returns to where it began).

Watch for & differentiate

Common slip: ignoring the turn and predicting a line for a shape loop.

Support: have students underline the turn line (or note 'no turn') first.

Extend: ask the shape for range(6) with right(60) (a hexagon).

Success looks like

The student predicts the shape/line and side count from a loop (Ex 1–3) and reasons that a closed shape returns Bit to the start (Ex 4).